

Real Estate Prediction System - Test Plan

Document Information

Item	Information
Project Name	Real Estate Prediction System
Document	Test Plan
Version	1.0
Prepared By	Nguyen Thu Hue
Date	July 2026
Type	Manual Testing
Test Management	Jira Software

1. Introduction

1.1 Purpose

This Test Plan defines the testing strategy, scope, objectives, schedule, resources, and deliverables for the **Real Estate Prediction System**.

The purpose of testing is to ensure that the system meets functional requirements, provides accurate property price predictions, and delivers a reliable user experience before release.

1.2 Project Overview

The Real Estate Prediction System is a web application that allows users to:

- Register and log into the system
- Search for real estate properties
- Predict property prices using Machine Learning
- Manage user profiles
- Contact administrators and submit feedback

The application is developed using:

- Python
- Flask
- HTML/CSS/JavaScript
- CatBoost Machine Learning Model
- SQLite Database

2. Scope

In Scope

The following modules will be tested:

Module	Description
Authentication	Login, Register, Forgot Password, Logout
Property Prediction	Predict property prices
Property Search	Search and filter properties
User Profile	Update profile, change password, upload avatar

Module	Description
Contact & Feedback	Contact form and feedback submission

Out of Scope

The following items are excluded:

- Source code review
 - Performance testing
 - Security penetration testing
 - Load testing
 - Mobile application testing
 - API automation testing
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3. Test Objectives

The objectives are to verify that:

- All functional requirements are correctly implemented.
 - Users can successfully complete business workflows.
 - Input validation works correctly.
 - Error messages are displayed appropriately.
 - Machine Learning prediction feature returns valid results.
 - UI elements display correctly.
 - Regression issues are identified before release.
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4. Test Strategy

Testing Levels

- System Testing
- Functional Testing
- Regression Testing
- UI Testing
- Validation Testing
- Smoke Testing

Testing Technique

- Black-box Testing
- Boundary Value Analysis
- Equivalence Partitioning
- Error Guessing

Defect Tracking

All defects will be tracked using **Jira Software**.

Bug severity:

- Critical
- High
- Medium
- Low

5. Test Environment

Item	Value
Operating System	macOS Sequoia
Browser	Google Chrome 138
Database	SQLite
Programming Language	Python 3.x
Framework	Flask
Machine Learning	CatBoost
Version Control	Git & GitHub
Test Management	Jira Software

6. Test Schedule

Activity	Duration
Test Planning	1 day
Test Case Design	2 days
Test Execution	3 days
Bug Reporting	2 days
Regression Testing	1 day
Test Summary Report	1 day

7. Entry Criteria

Testing will begin when:

- Functional requirements are completed.
- Application is deployed successfully.
- Test environment is available.
- Test data has been prepared.
- Build is stable.

8. Exit Criteria

Testing will be completed when:

- All planned test cases have been executed.
- No Critical defects remain open.
- High severity defects are resolved.
- Regression testing passes.
- Test Summary Report is completed.

9. Risks

Risk	Mitigation
Requirement changes	Update test cases immediately
Limited testing time	Prioritize critical modules
ML prediction inconsistency	Validate with sample datasets
Browser compatibility	Test on latest Chrome
Missing test data	Prepare representative datasets

10. Deliverables

The following documents will be delivered:

- Test Plan
- Test Cases

- Bug Reports
- Jira Issue Tracking
- Screenshots
- Test Summary Report
- GitHub Testing Portfolio